Exercise 19.1

11) It is a rectangle whose area $= 3 \times 5 = 15$ unit^2

12) It is a rhombus whose area $= \frac{\overset{3}{\cancel{6}} \times 4}{\cancel{2}} = 12$ unit^2

13) $D = (2, 2)$, Area $= 25$ unit^2

14)

$A = (0, 0)$
$B = (6, 0)$
$C = (0, 4)$
$D = (6, 4)$

15)

$A = (-3, 0)$
$B = (3, 0)$
$D = (-3, 3)$
$C = (3, 3)$

16)

~~$h^2 = a^2 + (2a)^2$~~
~~$h = \sqrt{a^2 + 4a^2}$~~
~~$a^2 + h^2 = 4a^2$~~
~~$h^2 = (2a)^2 - a^2$~~
~~$h = 2a - a$~~

let height be h

so

$h^2 + a^2 = 4a^2$
$h^2 = 4a^2 - a^2$
$h^2 = 3a^2$
$h = \sqrt{3}a$

$O = (0, 0)$
$A = (\sqrt{3}, a)$
$B = (2a, 0)$

17) As length of $QR = 4$, $RP = 4$.

so

~~$4^2 + x^2 =$~~
~~$4 + x^2 = 16$~~
~~$x^2 = 16 - 4$~~
~~$x = \sqrt{16-4}$~~
~~$= 4 - 2$~~
~~$= 2$~~

$4^2 + x^2 = 4^2$
$\Rightarrow 4 + x^2 = 16$
$\Rightarrow x^2 = 16 - 4$
$\Rightarrow x^2 = 12$
$\Rightarrow x = \sqrt{12}$
$= 2\sqrt{3}$

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